

Answers learning task 8

2026 – 2027

Exercise 5.13: Determine the value of c such that the function $f(x, y) = cxy$ for $0 < x < 3$ and $0 < y < 3$ satisfies the properties of a joint probability density function. Determine the following:

- (a) $P(X < 2, Y < 3)$
- (b) $P(X < 2.5)$
- (c) $P(1 < Y < 2.5)$
- (d) $P(X > 1.9, 1 < Y < 2.5)$
- (e) $E(X)$
- (f) $P(X < 0, Y < 4)$
- (g) Marginal probability distribution of the random variable X
- (h) Conditional probability distribution of Y given that $X = 1.5$
- (i) $E(Y|X = 1.5)$
- (j) $P(Y < 2|X = 1.5)$
- (k) Conditional probability distribution of X given that $Y = 2$

Answer

$$c = \frac{4}{81}$$

- (a) 0.4444
- (b) 0.6944
- (c) 0.5833
- (d) 0.3494
- (e) 2
- (f) 0
- (g) $f_X(x) = \frac{2}{9}x$, for $0 < x < 3$
- (h) $f_{Y|1.5}(y) = \frac{2}{9}y$, for $0 < y < 3$
- (i) 2
- (j) $4/9$
- (k) $f_{X|2}(x) = \frac{2}{9}x$, for $0 < x < 3$

Exercise 5.15: Determine the value of c that makes the function $f(x, y) = cxy$ a joint probability density function over the range $0 < x < 3$ and $0 < y < x$. Determine the following:

- (a) $P(X < 1, Y < 2)$
- (b) $P(1 < X < 2)$
- (c) $P(Y > 1)$
- (d) $P(X < 2, Y < 2)$
- (e) $E(X)$
- (f) $E(Y)$
- (g) Marginal probability distribution of X
- (h) Conditional probability distribution of Y given $X = 1$
- (i) $E(Y|X = 1)$
- (j) $P(Y > 2|X = 1)$
- (k) Conditional probability distribution of X given that $Y = 2$

Answer

$$c = \frac{8}{81}$$

- (a) $\frac{1}{81}$ (b) $\frac{5}{27}$
 (c) 0.7901 (d) $\frac{16}{81}$
 (e) 2.4 (f) 1.6
 (g) $f(x) = \frac{4}{81}x^3$, for $0 < x < 3$
 (h) $f_{Y|1}(y) = 2y$, for $0 < y < 1$
 (i) 0.6667 (j) 0
 (k) $f_{X|2}(x) = \frac{2}{5}x$, for $2 < x < 3$

Exercise 5.27: The weights of adobe bricks used for construction are normally distributed with a mean of 1.5 kg and a standard deviation of 0.3 kg. Assume that the weights of the bricks are independent and that a random sample of 20 bricks is selected.

- (a) What is the probability that all the bricks in the sample exceed 1.2 kg?
- (b) What is the probability that the heaviest brick in the sample exceeds 1.8 kg?

Answer

- (a) 0.032
- (b) 0.9684

Exercise 5.35: Determine the value for c and the covariance and correlation for the joint probability density function $f_{XY}(x, y) = cxy$ over the range $0 < x < 2$ and $0 < y < x$.

Answer

$$c = \frac{1}{2}, \quad \text{cov}(X, Y) = 0.0711, \quad \rho(X, Y) = 0.4924$$

Exercise 5.47: Suppose that X and Y have a bivariate normal distribution with $\sigma_X = 0.04$, $\sigma_Y = 0.08$, $\mu_X = 3.00$, $\mu_Y = 8.00$, and $\rho = 0$. Determine the following:

- (a) $P(2.95 < X < 3.05)$ (b) $P(7.60 < Y < 7.80)$
(c) $P(2.95 < X < 3.05, 7.60 < Y < 7.80)$

Answer

- (a) 0.7887 (b) 0.0062
(c) 0.0049

Exercise 5.49: In the manufacture of electroluminescent lamps, several different layers of ink are deposited onto a plastic substrate. The thickness of these layers is critical if specifications regarding the final color and intensity of light are to be met. Let X and Y denote the thickness of two different layers of ink. It is known that X is normally distributed with a mean of 0.1 millimeter and a standard deviation of 0.00031 millimeter, and Y is normally distributed with a mean of 0.23 millimeter and a standard deviation of 0.00017 millimeter. The value of ρ of these variables is equal to 0. Specifications call for a lamp to have a thickness of the ink corresponding to X in the range of 0.099535 and 0.100465 millimeter and Y in the range of 0.22966 and 0.23034 millimeter. What is the probability that a randomly selected lamp will conform to specifications?

Answer

0.8270